METRIC

MIL-S-24798(SH)
22 June 1995

MILITARY SPECIFICATION

SWITCH, POSITION, PROXIMITY (NON-CONTACT) OR LIMIT (MECHANICAL CONTACT), FIBER OPTIC, (METRIC), (NAVAL SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements and characteristics for fiber optic position switches (proximity or limit). This specification does not include switches which transfer an optical signal from one path to another by an external force or energy applied to the switch (see 6.1).

1.2 Classification (see 6.2).

1.2.1 Designations. The sensor classification shall consist of a series of designations which shall be assigned and listed in the format below.

24798	-	X	-	X	-	X
Specification	-	Type (see 1.2.1.1)	-	Opto-electronics module (see 1.2.1.2)	-	Electrical power supply (see 1.2.1.3)

1.2.1.1 Type.

Type I: Proximity Position Switch Sensor Head
Type II: Pin Actuated Limit Position Switch Sensor Head
Type III: Lever Actuated Limit Position Switch Sensor Head
Type IV: Lever and Roller Limit Position Switch Sensor Head
Type V: Tether Valve Limit Position Switch Sensor Head

Beneficial comments (recommendations, aditions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6021

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1.2.1.2 Opto-electronics module.

A - Bulkhead Mount (see 3.4.2.1)
B - Control Enclosure Mount (see 3.4.2.2)
C - Console Mount (Microprocessor expansion card) (see 3.4.2.3)

1.2.1.3 Electrical power supply. The optical and electrical interface wiring shall be determined by the power supply as follows.

AC - Four wire system used with a 115 Volt (nominal) alternating current (ac) supply.
DC -Four wire system used with a 28 Volt (nominal) direct current (dc) supply.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-901	-	Shock Test, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-17555	-	Electronic and Electrical Equipment Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-E-24142	-	Enclosures for Electrical Fittings and Fixtures, General Specification for.
MIL-E-24142/3	-	Enclosure for Electrical Fittings and Fixtures, Submersible, Size 6 by 9 (15-Foot).
MIL-M-24794	-	Material, Index Matching, Fiber Optics.
MIL-F-49291	-	Fiber, Optical,(Metric), General Specification for.
MIL-C-83522	-	Connectors, Fiber Optic, Single Terminus General Specification for.
MIL-C-83522/16	-	Connector, Fiber Optic, Single Terminus, Plug, Adapter Style, 2.5 Millimeters Bayonet Coupling, Epoxy.
MIL-C-83522/17	-	Connector, Fiber Optic, Single Terminus, Adapter, 2.5 Millimeter Bayonet Coupling, Bulkhead Panel Mount.

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MIL-C-83522/18	-	Connector Fiber Optic, Single Terminus, Adapter, 2.5 Millimeter Bayonet Coupling PC Mount.
MIL-C-85045	-	Cables, Fiber Optic, (Metric), General Specification for.

STANDARDS

MILITARY

MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Emission and Susceptibility.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1861	-	Electrical and Electronic Assemblies, Boards, Cards, and Associated Hardware, Selection and Use of.
MIL-STD-2042	-	Fiber Optic Topology Installation, Standard Methods for Naval Ships.

HANDBOOKS

MILITARY

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

455-20 - FOTP-20 Measurement of Change in Optical Transmittance (DOD adopted)
455-22 - FOTP-22 Ambient Light Susceptibility of Fiber Optic components.
455-34 - FOTP-34 Interconnection Device Insertion Loss Test.

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Reliability. The reliability of the opto-electronics module, expressed in mean time between failures (MTBF) shall be no less than 40,000 hours. A reliability prediction for the opto-electronics module, in the form of a MTBF calculation, shall be performed. The computational techniques specified in MIL-HDBK-217 should be used for guidance.

3.2 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) prior to production, in accordance with 4.4.

3.3 Materials.

3.3.1 Metals. Unless otherwise specified herein, all metals used in the construction of the proximity or limit position switch shall be corrosion-resistant and shall be in accordance with MIL-STD-454, requirement 15. Dissimilar metals shall not be used in intimate contact with each other unless suitably finished to prevent electrolytic corrosion and shall be in accordance with MIL-STD-454, requirement 16.

3.3.2 Flammable materials. Materials used in the construction of the proximity or limit position switch shall be noncombustible or fire retardant in the most hazardous conditions of atmosphere, pressure, and temperature to be expected in the application. Fire retardant additives may be used provided they do not adversely affect the specified performance requirements of the basic materials. Fire retardance shall not be achieved by use of nonpermanent additives to the basic material.

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3.3.3 Fungus resistant materials. Materials used in construction of the switch sensor head and its opto-electronics module shall be in accordance with MIL-STD-454, requirement 4, for fungus inert materials. Materials that are not in accordance with requirement 4 of MIL-STD-454 shall be in accordance with grade I classification of MIL-STD-810, method 508.

3.3.4 Solvents, adhesives, and cleaning agents. If any chemicals or cements are used in bonding of internal proximity or limit position switch components, no degradation shall result during in-service use. MIL-STD-454, requirement 23, may be used as guidance for epoxy and adhesives used in construction of the sensor.

3.3.5 Refractive index matching gels, fluids, or compounds. Refractive index matching gels, fluids, or compounds in accordance with MIL-M-24794 may be used.

3.4 Design and construction. The switch shall consist of an assembly with three major components: optical sensor head (Types I through V), fiber optic cables (see 3.4.3), and opto-electronics module (see 3.4.2). The opto-electronics module shall be interchangeable between any of the sensor types (I through V).

3.4.1 Sensor head. The design and construction of the sensor head (types I through V) shall be as specified herein.

3.4.2 Opto-electronics module. The design and construction of the opto-electronics module shall be as specified herein.

3.4.2.1 Bulkhead mounted (Type A). Bulkhead mounted opto-electronics modules shall be housed in a junction box in accordance with MIL-E-24142 and MIL-E-24142/3. The opto-electronics module inside the junction box shall be subjected to all first article tests as specified (see 3.2).

3.4.2.2 Control enclosure mounted (Type B). Control enclosure mounted opto-electronics modules are intended for use within an environmental protective enclosure, as part of a motor controller or other system. The opto-electronics module shall be subjected to all First Article tests as specified (see 3.2). The opto-electronics module shall be mounted in an enclosure as specified (see 6.2).

3.4.2.3 Console mounted (Type C). Console mounted (microprocessor or programmable logic controller (PLC) expansion card) opto-electronics modules are intended for use as a plug in card for a console control system. The size, weight, pinout configuration, and number of channels, shall be as specified (see 6.2). The opto-electronics shall be packaged in a console mounted circuit card as specified (see 6.2).

3.4.3 Fiber optic cable. For integrity, the cable shall have an outer diameter of a 4-fiber cable in accordance with MIL-C-85045. In the cable, there shall be a minimum of two times the number of fibers needed for

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operation of the sensor. Exposed single fiber Optical Fiber Cable Component (OFCC) shall not be used over distances greater than 1 meter, and in no case shall an OFCC penetrate a bulkhead, deck, or enclosure, except as an integral part of a cable.

3.4.4 Optical fiber. All optical fiber used in the construction of the fiber optic switch shall be in accordance with MIL-F-49291.

3.4.5 Fiber optic connectors, receptacles and bulkhead adapters. All fiber optic connectors, receptacles, and bulkhead adapters shall be in accordance with MIL-C-83522 and MIL-C-83522/16, 17, and 18, respectively, and shall be compatible for use with fiber optic cable in accordance with 3.4.3. Connectors shall be assembled at both ends of the fiber optic cable between the sensor head and the opto-electronics module. MIL-STD-2042 method 5B1 may be used as guidance.

3.4.6 Local status indication. The switch opto-electronics module shall have three indicator light emitting diodes (LEDs). A green LED which when lit indicates a switch closed indication, a red LED which when lit indicates a switch open condition, and a yellow LED which indicates a switch closed with alarm level condition. The LEDs shall be located on either the top or front of the module as it would be mounted during usage. The LEDs shall be visible in fluorescent room lighting. One LED and only one LED shall be lit at all times when the opto-electronics unit is energized.

3.4.7 Low intensity alarm set point adjustment. The sensor shall provide an indication of a degradation in the intensity of the sensor's optical signal via an alarm output. The opto-electronics module shall provide a means for adjusting the low intensity alarm set point by one individual and without the necessity for any electrical disconnection. The low intensity alarm set point adjustments shall be labeled and shall be accessible when the opto-electronics enclosure cover (for mounting configuration A) is removed. The low intensity alarm level set point shall allow tamper-proof sensitivity adjustment over the entire dynamic range of the opto-electronics module. The opto-electronics low intensity alarm will allow for an indication that maintenance is required, prior to a false open switch indication.

3.4.8 Lubrication. Lubrication of the opto-electronics module or the sensor head shall not be required.

3.4.9 Printed circuit boards. Printed circuit boards shall be in accordance with requirement 17 of MIL-STD-1861.

3.4.10 Fuses. The opto-electronics module shall not be fused.

3.4.11 Weight. The weight of the switch shall not be greater than 4.5 kilograms.

3.4.12 Dimensions. The critical dimensions of the switch shall be as specified (see 6.2).

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3.5 Switch electrical characteristics.

3.5.1 AC switch electrical characteristics. AC switches shall be two-wire devices and shall operate in series with the load.

3.5.1.1 Operating voltage. Nominal steady-state power supply requirements shall be 115 ± 8 Vac, 60 ± 2 Hz, single phase. Full time surge protection shall be provided for power supply limits (see 4.7.1).

3.5.1.2 Voltage drop. The voltage drop across each switch, during the activated (ON) state, shall not be greater than 4.0 Vac root mean square (rms) at rated load current (see 4.7.1).

3.5.1.3 Leakage current. The leakage current through each switch, during the deactivated (OFF) state, shall not be greater than 2.0 milliamperes (mA) rms (see 4.7.1).

3.5.1.4 Load ratings. The switches shall operate as specified with the load ratings of the switch as specified in 3.5.1.4.1 through 3.5.1.4.4.

3.5.1.4.1 Resistive. Switches shall have a resistive rating of 1.25.

3.5.1.4.2 Inductive. Switches shall operate inductive loads with a power factor between 1 and 0.35. As a minimum, switches shall have a make-current rating of 10.0 amperes rms for three cycles of the specified operating voltage and a break current rating of 1.25 amperes rms.

3.5.1.4.3 Minimum load. Switches shall operate with a minimum load of 15 mA rms.

3.5.1.4.4 Overload. Absolute overload and short circuit protection are desired. As a minimum, switches shall be protected from continuous overloads up to 6.0 amperes rms. Interruption of the operating voltage shall be required to restore normal operation of the switch after an overload has been detected.

3.5.1.5 Multiple AC switch operation. Multiple AC switch operation shall be as specified in 3.5.1.5.1 and 3.5.1.5.2 (see 4.7.1).

3.5.1.5.1 Series connection. Two AC switches of the same designation (see 1.2.1) shall operate in series (see 4.7.1). The total switch voltage drop at the load shall not be greater than 8.0 Vac rms at the rated load current.

3.5.1.5.2 Parallel connection. Two AC switches of the same designation (see 1.2.1) shall operate in parallel (see 4.7.1). The total switch leakage current at the load shall not be greater than 4.0 mA rms.

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3.5.1.6 AC wire colors. Wire colors shall be as follows:

Normally open (NO) switches:	Normally closed (NC) switches:
Black-Input	Black-Input
White-Output	White-Output

3.5.2 DC switches, electrical characteristics. DC switches shall be four-wire devices and shall operate as voltage sources. Switches shall be line powered and shall have solid-state outputs. Each switch shall have one normally open (NO) and one normally closed (NC) output in a complementary configuration. The total power drawn from the line shall not be greater than the sum of 50 mA plus the output load current.

3.5.2.1 Operating voltage. The nominal steady-state power supply requirements for dc shall be 28 ± 4.5V. Full time surge protection shall be provided for the power supply limits (see 4.7.1).

3.5.2.2 Voltage drop. The voltage drop across each switch, during the activated (ON) state, shall not be greater than 1.5 Vdc at rated load current (see 4.7.1).

3.5.2.3 Leakage current. Leakage current from each output in the open (OFF) state shall be not greater than 300 microamperes with 35 Vdc applied to the switch (see 4.7.1).

3.5.2.4 Load ratings. The switch shall operate as specified with the load ratings of the switch as specified in 3.5.2.4.1 an 3.5.2.4.2 (see 4.7.1).

3.5.2.4.1 Maximum current. The switches shall operate continuously and supply 250 mA to resistive and inductive loads and 100 mA into lamp loads. The inductive load shall have a decay time of not greater than 100 milliseconds when the load is interrupted. This decay shall be measured from the 90 percent level to the 10 percent level.

3.5.2.4.2 Minimum load. The switches shall supply any load current from maximum (see 3.5.2.4.1) down to zero.

3.5.2.5 Overload. The switches shall be protected from damage due to any overload or direct short circuit of the output. Interruption of the operating voltage shall be required to restore normal operation of the switch after an overload has been detected.

3.5.2.6 DC cable wiring. Wire colors shall be as follows:

Black - DC power high (positive lead)
White - NO output
Red - NC output
Green - DC power low (negative lead)

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3.5.2.7 Multiple DC switch operation. Multiple DC switch operation shall be as specified in 3.5.2.7.1 and 3.5.2.7.2 (see 4.7.1).

3.5.2.7.1 Series connection. Two DC switches of the same designation (see 1.2.1) shall operate in series (see 4.7.1). The total switch voltage drop at the load shall not be greater than 3.0 Vdc at the rated load current.

3.5.2.7.2 Parallel connection. Two DC switches of the same designation (see 1.2.1) shall operate in parallel (see 4.7.1). The total switch leakage current at the load shall not be greater than 600 microamperes.

3.6 Performance requirements.

3.6.1 Operation. The switch shall operate as specified in 3.4.6 and 3.5 (see 4.7.1).

3.6.2 Response time. Response time is the time it takes to go from 10 percent to 90 percent of full rise. The response time of the opto-electronics module shall be no greater than 100 milliseconds closed-open, 100 milliseconds open-closed (see 4.7.2).

3.6.3 Warm-up time. The switch shall operate properly as specified in 3.4.6 and 3.5 within 1 minute (see 4.7.3).

3.6.4 Change in optical transmittance. The change in optical transmittance shall not be greater than 3.0 dB during or after the change in optical transmittance test (see 4.7.4).

3.6.5 Dynamic range. The dynamic range of the opto-electronics shall be not less than 30 dB (see 4.7.5).

3.6.6 Ambient light susceptibility. The switch shall not indicate a false closed condition when the switch is open. The switch shall not indicate a false open condition when the switch is closed (see 4.7.6).

3.6.7 Steady-state voltage and frequency (ac) or supply voltage (dc). The switch shall not yield a false open condition when the switch is closed, nor shall it indicate a false closed condition when the switch is open during a steady-state voltage and frequency test (see 4.7.7). After the steady-state voltage and frequency test, the switch shall operate according to 3.6.1.

3.6.8 Transient voltage and frequency (ac) or voltage (dc). The output of the sensor shall not indicate an open switch condition at any time during the voltage of the transient voltage test (see 4.7.8). After the transient voltage and frequency, the switch shall operate in accordance with 3.6.1.

3.6.9 Insulation resistance. The insulation resistance of the opto-electronics module shall not be less than 10 megohms (see 4.7.9). After the insulation resistance test, the switch shall operate in accordance with 3.6.1.

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3.6.10 Power interruption. The output of the sensor shall not indicate an open switch condition during steady-state operation during the power interruption test (see 4.7.10). After the power interruption test, the switch shall operate in accordance with 3.6.1.

3.6.11 Short circuit. The local status indication shall indicate the correct switch position during the short circuit test (see 4.7.11). After the short circuit test, the switch shall operate in accordance with 3.6.1. The opto-electronics module shall exhibit no damage during a visual examination after the short circuit test.

3.6.12 Line voltage reversal (dc powered). The sensor shall operate in accordance with 3.6.1, after the line voltage reversal test (see 4.7.12). There shall be no evidence of visual damage to the opto-electronics module as a result of the line voltage reversal test.

3.6.13 Mechanical life. The sensor shall operate in accordance with 3.6.1 throughout the mechanical life test (see 4.7.13). The sensor shall show no evidence of physical damage during a visual inspection after the mechanical life test. The change in optical transmittance of the sensor shall remain within the limits specified in 3.6.4 after the mechanical life test.

3.6.14 Temperature. The sensor head and opto electronics module shall operate within the optical limits specified in 3.6.4 during and after the temperature test (see 4.7.14). The sensor shall show no evidence of physical damage during a visual inspection after the operating temperature test. After the temperature test, the switch shall operate in accordance with 3.6.1.

3.6.15 Humidity. The sensor head and opto-electronics module shall operate within the optical limits specified in 3.6.4 throughout the humidity test (see 4.7.15). The sensor shall show no evidence of deterioration of component parts or constituent materials, loosening of finishes, physical distortion, corrosion of metals, entrapment of moisture, separation of bonded surfaces or other damage during a post humidity test visual inspection. The output of the sensor shall not indicate an open switch condition at any time during the humidity test. After the humidity test, performance shall conform to 3.6.1.

3.6.16 Enclosure. There shall be no evidence of water leakage into the opto-electronics module or sensor head during an enclosure test (see 4.7.16). After the enclosure test, performance shall conform to 3.6.1 and 3.6.4.

3.6.17 Salt spray. The sensor head, opto-electronics module, components and connections shall show no visual evidence of deterioration such as flaking, pitting, blistering or loosening of finishes, corrosion of metal surfaces, or in the case of plated metals, corrosion which has passed through the plating and attacked the base metal during a visual examination after the salt spray test (see 4.7.17). No appreciable corrosion or other damage either optical, mechanical, or electrical, shall be evident after the salt spray test. After completion of the test, performance shall be in accordance with 3.6.1 and 3.6.4.

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3.6.18 Vibration. The sensor head and opto-electronics module shall meet the requirements specified herein during a vibration test (see 4.7.18). The sensor shall show no evidence of physical damage during a visual examination after the vibration test. The sensor shall operate within the optical limits specified in 3.6.4 after the vibration test. The output of the sensor shall not indicate an open switch condition at any time during the vibration test. At the completion of the test, performance shall conform to 3.6.1.

3.6.19 Shock. The sensor shall operate within the optical limits specified in 3.6.4 after each blow of the shock test (see 4.7.19). The sensor shall not pose a hazard to personnel or equipment due to major damage from shock blows. Minor deformation of the switch is acceptable, provided the sensor operates in accordance with 3.6.1 after the shock test. The switch shall not indicate an open switch condition for greater than 50 milliseconds during the shock test.

3.6.20 Electromagnetic interference (EMI). The output of the sensor shall not indicate an open switch condition at any time during the EMI test (see 4.7.20). The sensor shall operate within the optical limits specified in 3.6.4 after the test. The fiber optic switch shall operate in accordance with 3.6.1 after the EMI test. The sensors shall meet the following requirements of MIL-STD-461: CE101, CE102, CS101, CS114, CS116, RE101, RE102, RS101, RS103 and RS105 (see 4.7.20).

3.7 Surface finishes. Surfaces of castings, forgings, molded parts, stampings, machined and welded parts shall be free of defects such as cracks, porosity, undercuts, voids, and gaps as well as sand, dirt, fins, sharp edges, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and edges shall be either rounded or beveled. There shall be no burn-through. There shall be no warpage or dimensional change due to heat from welding operation. There shall also be no damage to adjacent parts resulting from welding.

3.8 Identification marking. Sensors shall be permanently and legibly marked in accordance with MIL-STD-130. Marking shall be located on the top or front of the sensor head (as mounted in service), and on the top or front of the opto-electronics module (as mounted in service). The following minimum information shall be provided:

3.8.1 Opto-electronics module.

(a) Manufacturer's name.
(b) Manufacturer's serial number.
(c) Manufacturer's part number.
(d) Electrical output (see 1.2.1.3).
(e) National Stock Number (NSN)
(f) The words "FIBER OPTIC POSITION SWITCH".
(g) Technical manual number.
(h) Contract number.

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3.8.2 Sensor head.

(a) Manufacturer's name.
(b) Manufacturer's serial number.
(c) Manufacturer's part number.
(d) Government designation (see 1.2.1).
(e) National Stock Number (NSN)

3.9 Labeling. A visible label shall be affixed to the outside of the opto-electronics module cover and shall contain the following:

NOTICE
UNTERMINATED OPTICAL CONNECTIONS MAY EMIT LASER RADIATION
DO NOT VIEW BEAM WITH OPTICAL INSTRUMENTS
AND AVOID DIRECT EXPOSURE TO THE BEAM

A visible label shall be affixed to the sensor head and the inside of the opto-electronics module and shall contain the following:

WARNING
INVISIBLE LASER RADIATION
AVOID EXPOSURE TO THE BEAM

The labels shall be yellow lettering on a black background.

3.10 Workmanship. Workmanship of the sensors shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance

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to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First Article inspection (see 4.4)
(b) Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Except where the parameters are a variable, all tests shall be conducted at nominal values of voltage, current and load. Standard environmental conditions shall be established for conducting the tests in the specification and shall only be changed when they are the variables in the tests.

4.4 First article inspection. When a first article inspection is required (see 3.2, 6.2, and 6.5), samples shall be inspected as specified in table I. Failure of any sensor to meet the requirements of this specification shall be cause for rejection.

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TABLE I. First Article Inspection.

Inspection	Requirement	Test Method
Group I: three samples required.		
General examination	3.3-3.5 and 3.7-3.10.	4.6
Operation	3.6.1	4.7.1
Response time	3.6.2	4.7.2
Warm-up time	3.6.3	4.7.3
Change in optical transmittance	3.6.4	4.7.4
Dynamic range	3.6.5	4.7.5
Ambient light susceptibility	3.6.6	4.7.6
Steady-state supply voltage and frequency (ac) or supply voltage (dc)	3.6.7	4.7.7
Transient voltage and frequency (ac) or voltage (dc)	3.6.8	4.7.8
Insulation resistance	3.6.9	4.7.9
Power interruption	3.6.10	4.7.10
Short circuit	3.6.11	4.7.11
Line voltage reversal	3.6.12	4.7.12
Mechanical life	3.6.13	4.7.13
Temperature	3.6.14	4.7.14
Humidity	3.6.15	4.7.15
Enclosure	3.6.16	4.7.16
Salt spray	3.6.17	4.7.17
Vibration	3.6.18	4.7.18
Shock	3.6.19	4.7.19
Group II: one sample required		
General examination	3.3-3.5 and 3.7-3.10	4.6
Operation	3.6.1	4.7.1
EMI	3.6.20	4.7.20

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4.4.1 Sample size. A sample shall consist of a sensor head (type I through V), an opto-electronics module, associated connectors, bulkhead adapters, and connector receptacles. Four samples of the same inspection lot (see 4.4.2) shall be subjected to first article inspection. Each sample shall be supplied with the length of cable required for application (see 6.2) or 30 meters of fiber optic cable in accordance with 3.4.3, whichever is greater. (Note that two items will be tested at the same time: the opto-electronics module and the sensor head, whether proximity or limit. Prior testing of an opto-electronics module in conjunction with the testing of a different sensor head does not exclude the opto-electronics module from any of the testing requirements specified herein.) Three samples shall be subjected to the tests of Group I and one sample shall be subjected to the tests of Group II.

4.4.2 Inspection lot. An inspection lot shall consist of all fiber optic switches of the same classification (see 1.2), produced under essentially the same conditions, in the same facility from the same materials and offered for delivery at the same time.

4.4.3 Order of inspection. The samples shall be subjected to the inspections specified in table I in the order listed except that the steady-state supply voltage and frequency inspection (see 4.7.7) may be performed concurrently with the temperature inspection (see 4.7.14). Any other deviation in the test order must first be approved by NAVSEA.

4.4.4 Disposition of first article units. Samples subjected to first article inspection shall be considered consumed and nondeliverable as part of the contract. Final disposition of samples shall be as specified (see 6.2).

4.5 Quality conformance inspection.

4.5.1 Group A inspection. Each fiber optic switch in each lot offered for delivery shall be subjected to the group A inspection shown in table II. Failure of any sensor to meet the requirements of this specification shall be cause for rejection.

TABLE II. Group A Quality conformance Inspection.

Inspection	Requirement	Test Method
General examination	3.3-3.5 and 3.7-3.10	4.6
Operation	3.6.1	4.7.1
Response time	3.6.2	4.7.2
Dynamic range	3.6.5	4.7.5

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4.5.2 Group B inspection. Samples shall be selected in accordance with table III. Each sample shall be subjected to the group B inspection specified in table IV. Failure of any sensor to meet the requirements of this specification shall be cause for rejection.

TABLE III. Group B inspection sample size.

Lot or batch size	Sample size	Lot or batch size	Sample size
2 to 8	2	151 to 280	32
9 to 15	3	281 to 500	50
16 to 25	5	501 to 1200	80
26 to 50	8	1201 to 4200	125
51 to 90	13	4201+	3 per 100
91 to 150	20		

TABLE IV. Group B Quality Conformance Inspection.

Inspection	Requirement	Test method
Steady-state voltage and frequency (ac) or voltage(dc)	3.6.7	4.7.7
Insulation resistance	3.6.9	4.7.9
Temperature	3.6.14	4.7.14
Enclosure	3.6.16	4.7.16

4.6 General examination. Each fiber optic position switch shall be given a thorough examination to determine conformance to the requirements of this specification with respect to material, finish, workmanship, construction, assembly, dimensions, and identification marking. Examination shall be limited to the examinations that may be performed without disassembling the units. Examination shall also include a check of all adjustments (see 3.3 through 3.5 and 3.7 through 3.10).

4.7 Methods of inspection.

4.7.1 Operation. The operation of the switch and opto-electronics module shall be tested by manually opening and closing the switch for 10 cycles. Performance shall be in accordance with 3.6.1.

4.7.2 Response time. The response time shall be measured by opening and closing the switch via a beam gate, controlled by a function generator. The output of the sensor shall be connected to a calibrated optical oscilloscope, or a calibrated electrical oscilloscope as appropriate. The output from the

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function generator shall be connected to the monitoring oscilloscope and the traces shall be compared to determine the response time of the system. Performance shall be in accordance with 3.6.2.

4.7.3 Warm-up time. The switch shall be deenergized for a period of not less than 12 hours. The switch shall then be energized. The warm-up time is the elapsed time between the application of line power and the point at which the sensor output reaches the conditions specified in 3.6.3. Performance shall be in accordance with 3.6.3.

4.7.4 Change in optical transmittance. The change in optical transmittance of the sensor head shall be performed in accordance with EIA-455-20. Performance shall be in accordance with 3.6.4.

4.7.5 Dynamic range. A calibrated optical attenuator with two jumpers shall be tested for insertion loss in accordance with EIA-455-34. The attenuator shall then be connected to the transmitter and receiver via the two jumper cables. The attenuation shall be increased from 0 dB (plus insertion loss of attenuator and jumpers) to 50 dB (plus insertion loss of attenuator and jumpers). The switch shall be left in the open position. The dynamic range will be exceeded when the sensor indicates a closed switch. The dynamic range shall be in accordance with 3.6.5.

4.7.6 Ambient light susceptibility. The ambient light source and general test conditions shall be in accordance with EIA-455-22. The entire switch shall be placed in the beam of the light source and placed in a closed condition. After a period of 10 minutes, the switch shall be placed in an open condition by disconnecting the optical cable from the source. The sensor shall then be subjected to the ambient light source for an additional 10 minutes. Performance shall be in accordance with 3.6.6.

4.7.7 Steady-state voltage and frequency. The switch shall be operated for not less than 15 minutes and shall be manually opened and closed not less than 10 times in each of the conditions A through E (ac) or F through H (dc) as specified in table V. Each of the conditions (either A through E or F through H) shall be tested at 0°C, 25°C, and 60°C. This test may be performed in conjunction with the temperature test (see 4.7.14). The switch shall be allowed to stabilize at each testing temperature before the steady-state voltage and frequency test shall be performed. Performance shall be in accordance with 3.6.7.

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TABLE V. Steady-state voltage and frequency test conditions.

Condition	Voltage			Frequency		
	Lower limit	Normal	Upper limit	Lower limit	Normal	Upper limit
(ac)						
A		115			60	
B	107			58		
C			123	58		
D			123			62
E	107					62
(dc)						
F	23.5			---	---	---
G		28		---	---	---
H			32.5	---	---	---

4.7.8 Transient voltage and frequency (ac) or voltage (dc). The switch shall be tested to 4.7.8.1 (ac) or 4.7.8.2 (dc). The sensor shall be placed in a closed switch position and the output of the sensor shall be monitored throughout the test. Performance shall be in accordance with 3.6.8.

4.7.8.1 Transient voltage and frequency (ac).

(a) With the switch operating at steady-state voltage of 123 Vac, the voltage shall be increased to 138 Vac, and then decreased back to the steady-state voltage of 123 Vac in a 2 second period.
(b) With the switch operating at a steady-state voltage of 107 Vac, the voltage shall be decreased to 92 Vac, and then increased back to the steady-state voltage of 107 Vac in a 2 second period.
(c) With the switch operating at a steady-state frequency of 62 Hz, the frequency shall be increased to 63.5 Hz, and then decreased back to the steady-state frequency of 62 Hz in a 2 second period.
(d) With the switch operating at a steady-state frequency of 58 Hz, the frequency shall be decreased to 56.5 Hz, and then increased back to the steady-state frequency of 58 Hz in a 2 second period.

4.7.8.2 Transient voltage (dc).

(a) With the switch operating at a steady-state voltage of 32.5 V dc, the voltage shall be increased to 34.5 V dc, and then decreased back to the steady-state voltage of 32.5 V dc in a 2 second period.

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(b) With the switch operating at a steady-state voltage of 23.5 V dc, the voltage shall be decreased to 21.5 V dc, and then increased back to the steady-state voltage of 23.5 V dc in a 2 second period.

4.7.9 Insulation resistance. The insulation resistance of the opto-electronics module shall be determined by applying 50 V dc between electrical input and output circuits and between these circuits and ground. The temperature shall be 25 ± 5°C and the relative humidity shall be 50 ± 10 percent. The insulation resistance measurement shall be made immediately after a 2 minute period of uninterrupted test voltage application. If the indication of insulation resistance meets the specified requirement (see 3.6.9) and is steady or increasing, the test may be terminated before the end of the 2 minute period. Performance shall be in accordance with 3.6.9.

4.7.10 Power interruption. The sensor shall be placed in a closed switch position and the output of the sensor shall be monitored throughout the test. With the switch operating within the steady-state tolerances of voltage and frequency, the external power supply shall be suddenly interrupted, and after an interval between 3 and 4 seconds, the power supply shall be reestablished to within the steady-state tolerances. After the switch has been operated long enough to detect any major performance degradation, the power shall be interrupted for an interval of 30 seconds. This cycle, (3 to 4 second interruption, monitor, then 30 second interruption) shall be repeated three times (total of 4). Performance shall be in accordance with 3.6.10

4.7.11 Short circuit. The switch shall be deenergized. The electrical output leads or terminals of the opto-electronics module shall be connected directly together, with no load resistance. The switch shall be energized for five minutes. The switch shall be manually opened and closed 10 times during this period. Immediately following the five minute period, the output pins shall be unshorted. Performance shall be in accordance with 3.6.11.

4.7.12 Line voltage reversal (dc). The switch power supply shall be connected as follows: the positive 28 V dc signal shall be applied to connector pin "B". The dc reference signal shall be applied to connector "A". The power supply shall be energized for a period of 10 minutes, and shall then be disconnected. The power supply shall then be correctly applied (pin A positive, pin B negative). Performance shall be in accordance with 3.6.12.

4.7.13 Mechanical life. The sensor shall be placed in closed condition. The sensor shall then be operated for 260,000 cycles (open-close is one cycle). The cycle rate shall be between one cycle per second and one cycle every two seconds (1 Hz to 0.5 Hz). A change in optical transmittance test (see 4.7.4) shall be performed at the end of the mechanical life test. Performance shall be in accordance with 3.6.13.

4.7.14 Temperature. The mated cable to sensor assemblies shall be tested at high and low temperature as specified herein. The switch shall be placed in a closed switch position and steps 1 through 6 shall be performed. The switch shall then be placed in an open switch position and steps 1 through 6 shall be

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performed. Change in optical transmittance shall be measured in accordance with 4.7.4. Performance shall be in accordance with 3.6.14. Visual inspection shall be performed after the test.

Step 1. Hold temperature at room ambient (25 ± 2°C) for one hour.

Step 2. Decrease temperature in steps of 10°C at 30 minutes per step until -28 ± 2°C is achieved.

Step 3. Hold temperature at -28 ± 2°C for 24 hours.

Step 4. Increase temperature in steps of 10°C at 30 minutes per step until +65 ± 2°C is achieved.

Step 5. Hold temperature at +65 ± 2°C for 24 hours.

Step 6. Decrease temperature in steps of 10°C at 30 minutes per step until 25 ± 2°C is achieved.

4.7.15 Humidity. The fiber optic position switch (sensor head, cable, and opto-electronics module) shall be subjected to the conditioning and tests specified in steps 1 through 4. The sensor shall be placed in a closed switch position and the output of the sensor (electrical and LED) shall be monitored throughout the test. Change in optical transmittance shall be measured in accordance with 4.7.4. Visual inspection shall be performed after the test. Performance shall be in accordance with 3.6.15.

Step 1. The fiber optic position switch shall be dried at a temperature not less than 40°C or more than 50°C for not less than 2 hours.

Step 2. Change in optical transmittance (see 4.7.4) shall be measured.

Step 3. The fiber optic position switch shall be subjected to four 24 hour cycles of temperature variation consisting of 18 hours at 65 ± 5°C and 6 hours at 25 ± 5°C. The rate of temperature change shall be no greater than 10°C per 30 minutes. The relative humidity shall be maintained at 95 percent during the steady-state conditions.

Step 4. Change in optical transmittance (see 4.7.4) shall be measured.

4.7.16 Enclosure. The fiber optic position switch (opto-electronics module, sensor head, and cable) shall be placed or mounted in the same manner that it will be secured on ship. The surface upon which the equipment is placed or mounted (supporting surface) shall extend not less than 1 meter beyond the equipment on all sides so that splashing may be produced by directing the water stream on that surface. The water flow shall be a coarse spray with a flow rate of not less than 60 liters per minute and a head pressure of not less than 3 meters. A head pressure of 3 meters is defined as sufficient water pressure so that if directed straight up, the stream of water shall rise

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to a height of 3 meters. The distance from the nozzle to the switch shall be approximately 2 meters. The time of the test shall not be less than 5 minutes with approximately equal portions of time for spray on each surface, including joints of the enclosure and at the supporting surface. Change in optical transmittance shall be measured in accordance with 4.7.4. Performance shall be in accordance with 3.6.16. Visual inspection shall be performed after the test.

4.7.17 Salt spray. The fiber optic position switch shall be tested in accordance with MIL-STD-202, method 101. The duration of the test shall be 96 hours. The sensor head, fiber optic cable, and opto-electronics module shall be placed in the chamber. The sensor shall not be energized during the test. Change in optical transmittance shall be measured in accordance with 4.7.4. Performance shall be in accordance with 3.6.17. Visual inspection shall be performed after the test.

4.7.18 Vibration. The sensor shall be placed in a closed switch position and the output of the sensor shall be monitored throughout the test. The sensor shall be tested in accordance with MIL-STD-167-1 type I vibration test. Change in optical transmittance shall be measured in accordance with 4.7.4. Visual inspection shall be performed after the test. Performance shall be in accordance with 3.6.18.

4.7.19 Shock. The sensor shall be subjected to the high-impact shock test for grade A, type A, class I, lightweight equipment as specified in MIL-S-901. The switch shall be placed in a closed switch position and the switch shall be monitored throughout the test. The change in optical transmittance shall be measured after each of the 9 hammer blows (see 4.7.4). Performance shall be in accordance with 3.6.19.

4.7.20 Electromagnetic effects. The sensors shall be tested in accordance with MIL-STD-462. The sensor shall be placed in a closed switch position and the output of the sensor shall be monitored throughout the test. Performance shall be in accordance with 3.6.20.

4.8 Inspection of packaging. Sample packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5.0 PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4).

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5.1 Preservation and packaging requirements. Fiber optic switches shall be preserved level A, C or commercial, packed level A, B, C, or commercial as specified (see 6.2) and marked in accordance with MIL-E-17555 and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2).

5.2 Marking. In addition to any special marking required (see 6.2), unit and intermediate packs and shipping containers and palletized unit loads shall be marked in accordance with ASTM-D-3951.

6. Notes.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Fiber optic proximity position switches are intended to be used in elevator motor controller systems, or other applications where a *noncontact position measurement method is desired. Fiber optic limit position* switches are intended for use as safety interlocks in a motor controller, as darken ship switches, as a contact type position indicator for an elevator control system, or other applications where a mechanical contact position *measurement method is desired. The fiber optic switches are intended to be* lightweight and inexpensive. The opto-electronics module may be either a stand alone unit, or an expansion card for a microprocessor based control system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of Fiber Optic switches required.
(c) Switch classification required (see 1.2).
(d) Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.2.1 and 2.2).
(e) When first article is required (see 3.2, 4.4, and 6.5).
(f) Opto-electronics modules, specify types A (see 3.4.2.1), B (see 3.4.2.2) or C (see 3.4.2.3). If type B or C, specify requirements.
(g) The critical dimensions of the switch (see 3.4.11).
(h) If responsibility for inspection equipment shall be other than as specified (see 4.1).
(i) Length of cable required (see 4.4.1).
(j) Final disposition of first article samples (see 4.4.4)
(k) Level of preservation and packaging required (see 5.1).
(l) Applicable acquisition requirements of MIL-E-17555 (see 5.1).
(m) Special marking required (see 5.2).

6.3 Data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific

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acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
Appendix	DI-DRPR-80651	Engineering drawings	Level 3
4.4	DI-NDTI-80809	Tests Inspection Reports	----

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards which have been cleared and listed in DOD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the sensors should be a preproduction sample, a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Subcontracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract. When ordering spare parts or repair parts for the equipment covered by this specification, the spare parts and repair parts should meet the same

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requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.8 Definitions.

6.8.1 Open Switch. The light path is blocked

6.8.2 Closed Switch. The light path is complete, signal from transmitter to receiver is complete.

6.8.3 Closed Switch with positive alarm. The light path is complete, signal level indicates that the end faces of the sensing element are dirty and require maintenance for continued proper operation.

6.8.4 Change in optical transmittance. The change in optical power level introduced by an environmental, mechanical or other induced stress.

6.8.5 Opto-electronics module. Unit contains the optical transmitter and receiver, signal conditioning electronics, signal output, and status indicating LEDs. May be an expansion card for a microprocessor based system, or a stand alone unit.

6.8.6 Sensor head. The opto-electronics module interrogates the sensor head to determine the position of the measurand. An optical signal is transmitted from the opto-electronics unit to the sensor head, the optical path is either complete or blocked depending on the status of the item being measured; giving an indication of the position or status of either an elevator, door, or hatch and so on.

6.9 Subject term (key word) listing.

Motor controller
Opto-electronics module
Safety interlock
Sensor head

Preparing activity:
Navy - SH
(Project 6021-N002)

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ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix contains the format and content preparation instructions for the development of and revision to a conformance verification drawing. It is not intended that each requirement contained herein should be applied to every type of instrumentation. Portions of this appendix are subject to deletion tailoring depending upon the material, construction, and principle of operating requirements that are specified in the individual instrumentation specification or acquisition document. This appendix is applicable only when data item description DI-DRPR-80651 is cited on the DD Form 1423. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY
MIL-I-45208 - Inspection System Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

20.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Y14.1 - Drawing Sheet Size and Format. (DOD adopted)
200 - Standard Reference Designations for Electrical and Electronics Parts and Equipments. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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30. CONFORMANCE VERIFICATION DRAWING FOR INSTRUMENTATION

30.1 Purpose. The conformance verification drawing contains the information necessary to verify that the instrumentation meets the requirements specified in the applicable instrumentation specification and acquisition document.

30.2 Content and format. One conformance verification drawing shall be developed for each specific type of instrumentation. It shall include all ranges, sizes, connections and other variations. The conformance verification drawing shall include the following minimum information (except as specified in 10.1) and shall be developed to the format in 30.3 through 30.9. A complete symbol list shall accompany all drawings and schematics.

30.3 Descriptive data.

(a) Instrument identification numbering system for instrumentation. This numbering system shall include, but may not be restricted to, the classification variables.
(b) Instrumentation identification number system for replaceable parts.
(c) Size, operating data, ranges, scale markings, and other data for proper selection.
(d) Test approval data, presenting the following information in tabular form:
    (1) Specification classification (unique alphanumeric variables to designate instruments).
    (2) Test report number and date.
    (3) Facility where test was conducted.
    (4) Authorized Government activity (NAVSEA) approval letter and date.
(e) A statement that instrumentation is in accordance with the requirements of the applicable instrumentation specification or acquisition document and to referenced specifications.
(f) Conformance verification drawing acceptance data, presenting the following information in tabular form:
    (1) Authorized Government activity (NAVSEA) acceptance letter and date.
    (2) Revision number.

30.4 Details of construction.

(a) Two or more representative assembly views, as required, to show clearly the details of the design, construction, and assembly of the instrumentation and to identify each part and its location. Identification of parts shall correspond to the list of materials. Assembly shall show how all mechanical parts are joined or attached.
(b) Sectional views or notes as necessary to show internal details.
(c) Details such as entrance provisions, gaskets, fastening techniques, welding symbols, mounting requirements, and other details as applicable.

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(d) Module enclosure requirements for separately mounted modules.
(e) Dimensions required to ensure interchangeability.
(f) Schematic of indicating system.
(g) Description of the nature and purpose of any adjustments.
(h) Welding procedures with acceptance data, including acceptance letter, report numbers and dates.
(i) Any special features.
(j) Complete weight.
(k) Location, size, and type of connections.
(l) Identification of quality control documents which show conformance with MIL-I-45208 or the quality control specification that is listed in the applicable instrumentation specification or acquisition document. The acceptance letter shall be referenced.

30.5 Parts list. The following information shall be presented in tabular form:

(a) Item number (corresponding to flag number identifying the part on the conformance verification drawing).
(b) Quantity of each part required per assembly.
(c) Name of part with sufficient information to readily identify the part (for example, screws: thread size, length and type of head, shall be specified).
(d) Material of part.
(e) Material specification (military, federal, or non-Government document number or Government activity (NAVSEA) drawing number).

NOTE: When substitution of a material specification is made, it is the responsibility of the contractor to provide written documentation to substantiate that the substituted material is equivalent to the specified material.

(f) Type, class, grade, size, military designation, or other classification of any referenced specification.
(g) Part number or identification assigned by assembly supplier.
(h) Name of actual manufacturer of part (when applicable).
(i) Part number or identification assigned by part supplier (when applicable).
(j) Onboard repair parts. Parts that are appropriate for, or are supplied as, onboard repair parts shall be indicated by a symbol in this column.
(k) Remarks column. Finishes, platings, or coatings along with the applicable specification or other requirements should be specified in this column.

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30.6 Table for special tools. Special tools required for the instrumentation shall be presented in the following tabular form:

(a) Item number (corresponding to flag number identifying the tool on the conformance verification drawing).
(b) Quantity of each tool required per instrument.
(c) Name (description) of tool (include generic name).
(d) Tool specification (military, federal, or non-Government document number or Government activity (NAVSEA) drawing number).
(e) Tool number or identification assigned by assembly supplier.
(f) Name of actual manufacturer of tool (when applicable).
(g) Tool number or identification assigned by tool supplier (when applicable).
(h) Description of tool's application.
(i) Remarks column. Special techniques or other usage requirements shall be explained in this column.

30.7 Electrical system schematics. Electrical schematics shall be included in the verification conformance drawing. A complete schematic shall contain all the parts in the electrical systems. A single schematic shall represent clearly the operation and the function of the electrical circuitry within the instrumentation. The schematic shall contain all parts (including components, connectors, alarms, and so forth) which make up the electrical system. When confusion over its function could result, a simplified schematic containing only the major components shall also be provided. This simplified schematic may be presented in block diagram format. The following features shall be incorporated into the schematic.

(a) In preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram.
(b) A thin, broken line shall be used to represent the boundaries of each unit or subassembly. Terminals, to which external connections are made, shall be shown within these boundaries, with the numbers, markings, type of signal, power and ground, as appropriate.
(c) Each part (such as resistors, capacitors, relays, and so forth) shall be given a unique reference designation consisting of a letter denoting the type of part (as required by ANSI 200) and a number assigned consecutively. The numbers shall be assigned in a logical sequence of electrical current or signal flow through the circuit.
(d) In addition to where the reference designation, parts not conforming to a military specification, where permitted, shall have the following information noted adjacent to the part. Where numerical values are given, a code shall be noted to designate the units used:
    (1) Resistors - resistance, power rating, and tolerances. If variable, an arrow to indicate clockwise rotation of the control shaft.
    (2) Capacitors - capacitance, voltage rating, and tolerance.

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(3) Reactors - inductance and voltage rating.
(4) Semi-conductors - type number (JEDDC number is adequate).
(5) Integrated circuits - operational symbol diagram of input-output relationship, terminal numbering corresponding to a representative schematic, and type number.

(e) Supply voltages, phases, and frequencies and transformer terminal voltage shall be indicated and labeled as to purpose.

(f) Table for troubleshooting of electrical system. The content table shall include each test point, as identified on the electrical system schematic, with voltage, waveform or other electrical parameter that should be measured at each test point.

30.8 Selection and installation considerations.

(a) Performance data.
(1) Accuracy (in percent of span).
(2) Shock and vibration classification.
(3) Degree of water tightness of the enclosure.
(4) Electromagnetic interference and pulse susceptibility.

(b) Dimensional outline of the instrumentation showing overall and principle dimensions in sufficient detail to establish space requirements in all directions necessary for installation, servicing, exclusive of space required for operator observation of the indication.

(c) Special considerations which may affect selection or installation.
(1) Ambient temperature range.
(2) Calibration points and adjustments.
(3) Orientation.
(4) Location of instrumentation relative to vibrating equipment.
(5) Protection of the instrumentation from pulsations and spikes in the parameter being measured.
(6) Selection of the instrumentation range relative to the operating range of the system.
(7) Application for each type connection.
(8) Cleaning procedure or reference to the cleaning procedure used.
(9) Selection of the instrumentation for compatibility (materials, temperature, pressure, and so forth) with the ambient environment and with the parameter being measured.

30.9 Drawing format.

(a) Unless otherwise approved by the authorized Government activity, a maximum of three sheets shall be allotted for single functioned system instrumentation and a maximum of ten sheets for instrumentation containing a multiple functioned system. A single functioned system is one that performs only one operation such as: expands and contracts a pressure elastic element, steps up and steps

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down the voltage, conditions one electrical signal, or winds and unwinds a bimetallic element. A multiple functioned system is one that contains two or more single functioned systems.

(b) Each sheet shall be zoned.

(c) Title block shall be included on each drawing sheet and shall include the following information:

(1) Title, drawing number, and revision letter. Each sheet shall contain the same title, drawing number, and revision letter.

a. Title - the title shall consist of the name by which the instrumentation is known.

b. Drawing number - the drawing number shall consist of alphanumeric characters which may be separated by dashes or slashes. The total number of characters in the drawing number (including dashes and slashes) shall not be greater than 15. Blank spaces are not permitted within the drawing number.

c. Revision letter - the revision letter shall denote the latest approved version of the drawing. The revision letter of conformance verification drawing shall not be changed until all the changes under that revision have been accepted in writing by the authorized Government activity (NAVSEA). No changes made to the conformance verification drawing shall be considered a revision until after the initial version of the conformance verification drawing has been accepted in writing by the authorized Government activity. After initial submittal of the conformance verification drawing, no changes shall be made during the initial drawing review process unless the change is either requested by the authorized Government activity or the change is documented in written correspondence by the contractor.

(2) Sheet _____ of _____.

(3) Tolerance on dimensions for fractions, decimals, and angles. Units of the dimensions specified on the conformance verification drawing.

(4) Contractor acceptance block (appropriate signatures and dates).

(5) Commercial and Government entity (CAGE) code for manufacturer.

(6) Scale.

(7) Reference drawings.

(8) Manufacturer's name and address.

(9) Drawing size.

(d) A revision block shall be included on each sheet of the conformance verification drawing and shall contain the following information in tabular form:

(1) Revision letter.

(2) Description of revision.

(3) Acceptance letter serial number and originator identification.

(4) Acceptance data.

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(e) Sheet size and format not specified herein shall be in accordance with ANSI Y14.1.
(f) Classification designations. No Government security classification designation such as confidential or secret shall appear on the conformance verification drawing unless a particular classification is specified by the Government.

40. CONFORMANCE VERIFICATION DRAWING ACCEPTANCE

40.1 Acceptance. Acceptance shall be granted by the authorized Government activity only after the conformance verification drawing is found to meet all the requirements specified in 20. through 40.1.

50. INSTRUCTIONS

50.1 Acquisition document instructions. The acquisition document should contain provisions that address submission, review, extension, disapproval, default, acceptance, and waiver of conformance verification drawings in addition to the effects on the delivery schedule due to delays in conformance verification drawing acceptance. NAVSEA shall be designated as the activity that accepts or disapproves the conformance verification drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-24798(SH)	95/06/22

3. DOCUMENT TITLE SWITCH, POSITION, PROXIMITY (NONCONTACT) OR LIMIT (MECHANCIAL CONTACT), FIBER OPTIC, (NAVAL SHIPBOARD USE) (METRIC)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY TECHNICAL POINT OF CONTACT (TOPC):

a. NAME	b. TELEPHONE (Include Area Code)	
Mr. Frank Leland SEA 03J4 ALL CORRESPONDENCE SHOULD BE ADDRESSED TO:	(1) Commercial (703)602-6137	(2) AUTOVON 332-6137
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command SEA 03R42, 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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